



CTEC 21063 - Database Systems

Faculty of Computing and Technology, University of Kelaniya

Group Assignment (4 members)

Learning Outcomes covered:

- LO-2 Design a relational database system using the ER and EER Model
- LO-3 Design a relational database using the normalization approach
- LO-5 query the database using SQL

*****This assessment contributes 15% to the Final grade.**

Tasks to be done:

Realizing the database design process

Requirement gathering

1. Select a business process that you know. Example: Food delivery system, Attendance management system, Student registration system etc. You must find a business for a real client.
2. Briefly describe your business process by identifying its main functions and its **data requirement** for the system of the selected business (100 - 200 words)

*****Above task 1 and 2 represent the requirement gathering stage.**

Design the database for the system

3. Model the Data requirements using the Entity Relationship (ER) Model.
4. Convert the ER Model to derive the Relational Data Model (Schema of the Database). There should be a minimum of 8 relations. (Each member should be responsible for at least two relations)
5. Write SQL commands (DDL) to create database tables (for the relations identified in the relational model).
6. Add necessary constraints to the relations (Example: Primary key, foreign keys, Unique, Check and not null etc.)
7. Write SQL commands (DML) to insert records for the created tables (minimum 4 for each table).
8. Each member should write at least 3 DML statements from their responsible relations to retrieve 3 different types of information useful to the business (There should be minimum 12 SQL statements from the group). You should demonstrate your capability in retrieving data by joining multiple tables, using aggregate function, Having, Order by etc.



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***Minimum requirements are mentioned above. But you may use your in-depth database knowledge to design the database and demonstrate it in the report.

Implementation

9. Create a database instance (name it as CTEC21063_G<your group number> E.g., CTEC21063_G1) using PostgreSQL. Run the DDL and DML commands accordingly and implement your designed database.
10. Collect the related data dictionaries for the created relations.
11. Create a database dump. (Refer to <https://www.postgresql.org/docs/current/backup-dump.html>)

PostgreSQL provides the utility program **pg_dump** for this purpose. The basic usage of this command is: `pg_dump your_db_name > dumpFileName`

Project Deliverables

1. A report (Word document) including the details of the Project Tasks that are mentioned above. Refer to the Appendix I below to get the cover sheet for the report. Include the declaration form (Appendix II) as the second page.

Report should include the following information.

- I. Title page
- II. Declaration form with all student IDs
- III. Table of Contents
- IV. List of Figures
- V. List of Tables
- VI. Introduction of the business with functional and data requirements (Task 2)
- VII. ER/EER Diagram with assumptions if any (Task 3)
- VIII. Relational model with proper arrows showing primary key – foreign key relationships (Task 4)
- IX. Data Definition Language (DDL) SQL statements (Tasks 5 and 6)
- X. Data Dictionaries for all relations (Task 10)
- XI. Data Manipulation Language (DML) SQL statements used to insert data (Task 7)
- XII. DML SQL statements used to query business information (Task 8)



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***For the above items from IX. to XII.), responsible student ID should be clearly mentioned.

2. Database dump file. (Examiners should be able to restore the dump for checking purpose. Therefore, check the restorability of your dump by importing it again in a different machine before submitting)

*** You must create a zip file of deliverables and upload it to the submission link provided in the LMS.

Submission deadline

9th November 2025 (On or before 11:59 pm)

IMPORTANT:

- Zero toleration for plagiarism. (If noticed, assign zero marks)
- No extensions for the deadline.



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Appendices

Appendix I: Cover Page for the report.



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Database Systems
[CTEC 21063]

Assignment – Realizing database design process
<Name of your project>

Date of submission: DD/MM/YYYY



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Appendix II: Declaration page.

Declaration

We hereby certify that this report of our own work and does not contain any material previously published or written by another person except where due reference is made in the text.

Student ID	Name	Signature

Date: dd/mm/yyyy



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Assignment Marking Guide:

- 1) Report **(90 marks)**
 - I. Described business scenario (10 marks)
 - a) Process functions identification
 - b) Data requirement identification (entities)
 - II. ER diagram (20 marks)
 - a. Entities [.5*8=4]
 - b. Attributes with Primary Key= [4]
 - c. Relationships[4]
 - d. Cardinality[5]
 - e. Completeness and richness ex: having Week entity, descriptive attributes etc.[3]
 - III. Relational model or Schema (20 marks)
 - a) Mapping strong entities correctly (.5*8=4 marks)
 - b) Mark the primary key (4marks)
 - c) Correct foreign key (6 marks)(include the FK in the correct location – 3, arrow with correct arrow head - 3)
 - d) New relation introduces for M:N relationship (3)(new relation -1, FK marking-1, PK marking -1)
 - e) Completeness (3 marks) (incorporating weak entity, descriptive attributes etc.)
 - IV. DDL statements with constraints (20 marks)
 - a) Create table statement(4)
 - b) Correctly adding PKs(4)
 - c) Correctly adding FKs(4)
 - d) Right domains (types of data)(4)
 - e) Adding other constraints and correctness (Check constraints, NOT NULL, DEFAULT etc.)(4)
 - V. DML statements – Insert (8 marks)
 - VI. DML statement – Select (12 marks)
- 2) Database dump **(10 marks)**

-----End of the assignment-----